

NITIN YADAV

Rewari, Haryana, India | nitin.y2309@gmail.com | +91-9244502740 | [GitHub](#) | [LinkedIn](#) | [LeetCode](#)

Professional Summary

Computer Science undergraduate at VIT Bhopal University (CGPA: 8.03/10) with hands-on experience building AI-powered applications using Python, FastAPI, Next.js, TensorFlow, and Gemini AI. Passionate about machine learning, computer vision, backend engineering, and developing scalable AI solutions.

Technical Skills

Languages: Python, C++, Java, SQL

AI / ML & LLMs: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, OpenCV, Gemini API, Prompt Engineering

Backend & APIs: FastAPI, Flask, REST API Integration

Frontend: HTML, CSS, JavaScript, Next.js, TypeScript

Database: MySQL

Tools: Git, GitHub, Google Colab, VS Code, Jupyter Notebook

Core: DSA, DBMS, OOP, Operating Systems, Probability & Statistics

Projects

MotivationAI — AI-Powered Customer Motivation Detection

Hackathon MVP

Next.js 15, TypeScript, FastAPI, Google Gemini

- Built an end-to-end full-stack platform classifying users into 6 psychographic motivation profiles in real time using Gemini AI.
- Designed an executive analytics dashboard featuring KPI cards, AI-generated executive summaries, signal explainability, and motivation-drift detection.
- Generated personalized, context-aware marketing recommendations with enriched recommendation cards to maximize actionability.
- Built a modular FastAPI backend with REST APIs for real-time inference and analytics.

Crop Disease Detection using Explainable AI

Academic Project

TensorFlow, Keras, OpenCV, Flask

- Developed a CNN-based plant disease classifier with image preprocessing and augmentation to improve model robustness.
- Integrated Grad-CAM visualization to provide explainable, interpretable predictions for end users.
- Built a Flask application for real-time disease prediction from uploaded images.

Battery Health Prediction System

Academic Project

Python, Scikit-learn, FastAPI

- Built a Random Forest classifier to predict vehicle battery health from voltage, current, temperature, age, and internal resistance.
- Developed a FastAPI backend with an interactive prediction dashboard for real-time inference.

Education

VIT Bhopal University

2023 – 2027

B.Tech in Computer Science and Engineering | CGPA: 8.03/10

M.L.P. Senior Secondary School, Rewari

2023

Class XII — PCM + Computer Science

Euro International School, Rewari

2021

Class X — CBSE

Certifications

- NPTEL Marketing Analytics — Elite (93%)
- NPTEL Introduction to Machine Learning
- Google IT Support Professional Certificate
- Coursera — Bits and Bytes of Computer Networking

Relevant Coursework

Machine Learning, Data Structures and Algorithms, Database Management Systems, Object-Oriented Programming, Operating Systems.